

INDUSTRIAL PRESSURE TRANSMITTER

For Process Industry



Company Profile

Shanghai LEEG Instruments Co., Ltd. (LEEG), a wholly-owned subsidiary of MICRO SENSOR CO., LTD., serves as the group's dedicated measurement and control solutions platform for the process industries. We are committed to providing highly reliable and high-performance pressure measurement and process monitoring solutions for complex and demanding industrial applications.

With over two decades of experience in the industry, we are now a leading company of smart pressure transmitter in China. We have accumulated extensive expertise in the R&D, manufacturing, and field application of monosilicon pressure transmitters, with cumulative product deliveries exceeding one million units. Our products are certified by internationally recognized standards such as CE, ATEX, IECEx, CSA, 3A, and EHEDG, and are widely used in oil and gas, chemical, new energy, municipal environmental protection, food and beverages, biopharmaceuticals, and high-end equipment manufacturing sectors. Our company operates a CNAS-accredited laboratory, a Class 100 cleanroom, and a digital factory. Supported by ERP, WMS, and MES systems along with automated equipment, we achieve full-process digital management and traceable information flow. With outstanding product quality and efficient delivery capabilities, we consistently set new benchmarks in the field of process measurement and control.

Manufacturing Base

📍 Shanghai, China



10,000 m²
Site Area

5,000 m²
Production Area

500,000 units/year
Annual Capacity

Product Overview

The LEEG monosilicon Pressure Transmitter combines monocrystalline-silicon compound strain sensing technology, modular design, full-range digital temperature compensation, and self-diagnostics. It offers strong immunity to interference, vibration resistance, and long-term stability. The transmitter measures pressure/differential pressure, level, density, and flow in liquids, steam, and gases. It supports multiple interfaces, including 4–20 mA + HART 7, Modbus RS-485, PROFIBUS-PA, and 1–5 V low-power, with complete process connections. Parameters can be set and monitored on-site via push buttons, handheld devices, or PC, providing reliable solutions for industrial process automation.

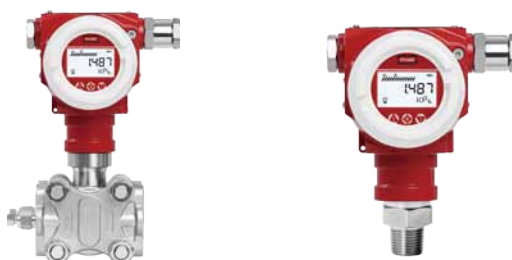
Features

- Over 1 million units installed worldwide
- Basic accuracy: $\pm 0.075\%$ FS; optional $\pm 0.05\%$ FS
- Wide selection of corrosion-resistant materials, suitable for corrosive, crystallizing, and high-temperature environments
- Safe and reliable explosion-proof design, certified to international standards



Product Series

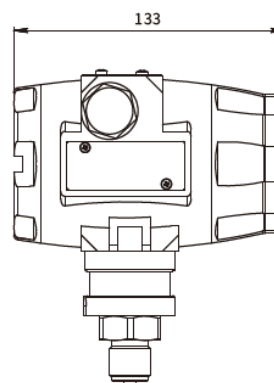
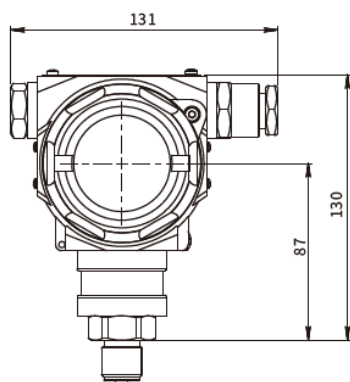
DMP305X Series



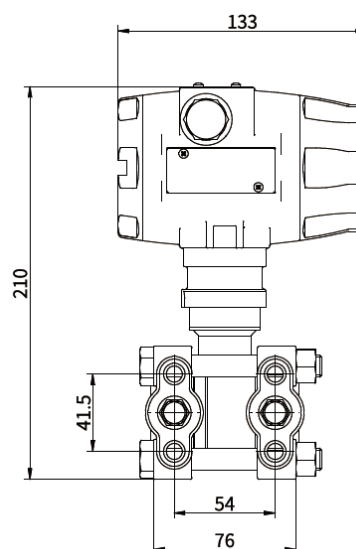
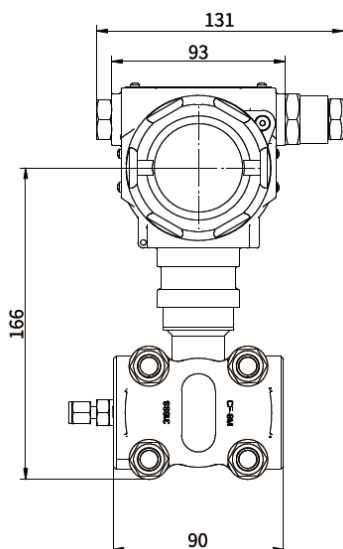
SMP858 Series



Dimensions



DMP305X-TST Smart Pressure Transmitter



DMP305X-DST Smart Differential Pressure Transmitter

Key Advantages

Accurate Measurement & High Reliability

- 1 Equipped with an advanced MEMS sensor chip and patented control module, offering full-range digital temperature compensation ($-40^{\circ}\text{C}\sim+85^{\circ}\text{C}$)
- 2 Dual-chamber housing enhances sealing and EMI immunity, with surge-protected terminals for added safety
- 3 Built-in self-diagnostics provide real-time status monitoring and fault detection to minimize downtime

Excellent Performance in Harsh Conditions

- 1 Vibration effect $<0.1\%$ (per GB/T18271.3 & IEC 61298-3) ensures stable operation under vibration
- 2 Potted key components improve corrosion and contamination resistance
- 3 Fully welded diaphragm system with multiple corrosion-resistant options

Fully Isolated Packaging

Provides temperature, media, and EMI isolation
Robust overpressure protection



Ultra-High Temperature Solution

Supports molten salt measurement up to 600°C for new energy applications



Comprehensive Diaphragm Seal System

Multiple corrosion-resistant diaphragm materials and treatments optional
Suitable for high-temperature, high-vacuum, and corrosive media



Representative Application Cases

High-concentration chlorine and mixed-acid off-gas
High-frequency vibration pressure measurement
Repeated vacuum-pressurization cycling
 $>500^{\circ}\text{C}$ ultra-high-temperature media

High-temperature HF vapor
Slurry media with solid particles
Adhesive or crystallizing media
High vacuum with corrosive gases

Performance Specifications

- Basic accuracy: $\pm 0.075\%$ FS; optional $\pm 0.05\%$ FS
- Measuring range: Differential pressure: 100Pa~10MPa; Pressure: 2kPa~40MPa
- Turn down (range ratio): 100:1
- Output signals: 4–20 mA, 4–20 mA + HART 7, 1–5 V low power, Modbus-RTU (RS485), PROFIBUS-PA
- Overpressure: 25 MPa (single-sided)
- Long-term stability: $\pm 0.1\%$ SPAN/5 years
- Diaphragm materials: 316L, HC-276, Monel 400, Ta, Ti, Duplex 2205, etc.
- Diaphragm surface treatments: Gold plating, PTFE coating, diamond-like coating, PFA protective cap, degreasing, etc.

【Static Pressure Effect】	
Nominal range	Typ. / 10 MPa
6kPa	$\leq \pm 0.5\%$ F.S.
40kPa	$\leq \pm 0.1\%$ F.S.
250kPa	$\leq \pm 0.075\%$ F.S.
1MPa	$\leq \pm 0.075\%$ F.S.
3MPa	$\leq \pm 0.15\%$ F.S.
10MPa	$\leq \pm 0.3\%$ F.S.

【Response Time】	
Damping time constant: equals to the combined damping time of electronic components and sensor module	
Electronic components damping: 0-100s adjustable	
Sensor module damping (isolated diaphragm and silicone oil filled): ≤ 0.2 s	
Turn-on time: < 2.5 S	
Data recovery time: ≤ 15 S	
Response: ≤ 100 ms	

【Mounting Position Effect】	
Less than 400Pa at any position, which can be corrected by PV (primary value)=0 reset	

【Vibration Effect】	
$< 0.1\%$ as per GB/T18271.3/IEC61298-3	
Test conditions: frequency 10–2000 Hz; displacement amplitude 0.21mm or 0.35mm; acceleration amplitude 29.4 or 49m/g ²	

【Insulation Resistance】	
All signals*	≥ 100 MΩ at 500V DC under reference conditions
* Insulation resistance applies only to power terminals; testing signal lines may damage the instrument.	

【Environmental Conditions】	
Item	Operating conditions
Ambient temperature	Without display: -50°C~85°C; With LCD display: -40°C/-20°C~70°C
Storage temperature	Without display: -50°C~100°C; With LCD display: -40°C/-20°C~85°C
Medium temperature	Standard silicone oil: -40°C~85°C
	Low-temp silicone oil: -55°C~85°C
	Fluorinated oil (2): -55°C~85°C
Ambient humidity	5-100%RH
IP rating	IP67

【EMC Environment】				
No.	Test item	Basic standards	Test conditions	Performance level
1	Radiated interference (Housing)	GB/T 9254/CISPR22	30MHz-1000MHz	Pass
2	Conducted interference (DC power port)	GB/T 9254/CISPR22	0.15MHz-30MHz	Pass
3	Electrostatic discharge (ESD) immunity test	GB/T 17626.2/IEC61000-4-2	4kV (contact), 8kV (air)	A (Note 1)
4	Radiated, radio-frequency, electromagnetic field immunity test	GB/T 17626.3/IEC61000-4-3	10V/m (80MHz-1GHz)	A (Note 1)
5	Power frequency magnetic field immunity	GB/T 17626.8/IEC61000-4-8	30A/m	A (Note 1)
6	Electrical fast transient/burst immunity test	GB/T 17626.4/IEC61000-4-4	2kV (5/50ns, 100kHz)	A (Note 1)
7	Surge immunity test	GB/T 17626.5/IEC61000-4-5 GB/T 17626.6/IEC61000-4-6	2kV (line-to-line) 4kV(line-to-ground) (1.2/50μs)	A (Note 1)
8	Immunity to conducted disturbances, induced by radio-frequency fields		3V (150kHz-80MHz)	A (Note 1)
Note 1: Performance level A — normal performance within technical specifications. Note 2: Measurement data above applies to output type “F, 4H” .				

Industry Applications



Oil & Gas

Measurement and control of wellhead pressure and level, storage tank pressure and level, and pipeline pressure



Chemical Industry

Measurement and control of reactor pressure and level, pipeline pressure, and raw material and finished product tank pressure and level



Biopharmaceuticals

Measurement and control of operating pressure in intermediate reactors, pressure and level in evaporators, and pressure and level in fermenters



New Energy

Measurement and control of pressure and level in hydrogen production systems, separator level, and kiln/furnace pressure in lithium battery production



Municipal & Environmental

Measurement and control of flue gas duct outlet pressure and wastewater/aeration tank level



Advanced Machinery & Equipment

Measurement and control of hydraulic system pressure, vacuum system pressure, and tank pressure and level



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